Creating a Custom MySQL Docker Image with Database Initialization

To Begin with the Lab

Summary of the Lab

This lab demonstrates how to create a custom MySQL Docker image with automatic database initialization using SQL scripts. It covers creating a Dockerfile, building and running a MySQL container, and verifying database tables and sample data. The lab also explains how to use Docker volumes to persist MySQL data across container removal and recreation. Finally, it shows how to update the SQL initialization script, rebuild a new MySQL image, recreate the volume, deploy a fresh container, and confirm that the customized database, users, and data are initialized correctly.

- Create a Project Directory

```
mkdir mysql-custom-image

cd mysql-custom-image
```

- Create a file named **01.sql**.

```
nano 01.sql
```

- Add the following SQL script (using a different example):

```
CREATE TABLE employees (

    employee_id INT AUTO_INCREMENT PRIMARY KEY,

    employee_name VARCHAR(100),

    department VARCHAR(50),

    salary DECIMAL(10,2)

);

INSERT INTO employees (employee_name, department, salary)

VALUES

('Aman','Cloud',65000),

('Rahul','DevOps',72000),

('Priya','AI/ML',85000);

CREATE USER 'appuser'@'%' IDENTIFIED BY 'AppUser@123';

GRANT ALL PRIVILEGES ON company_db.* TO 'appuser'@'%';
```

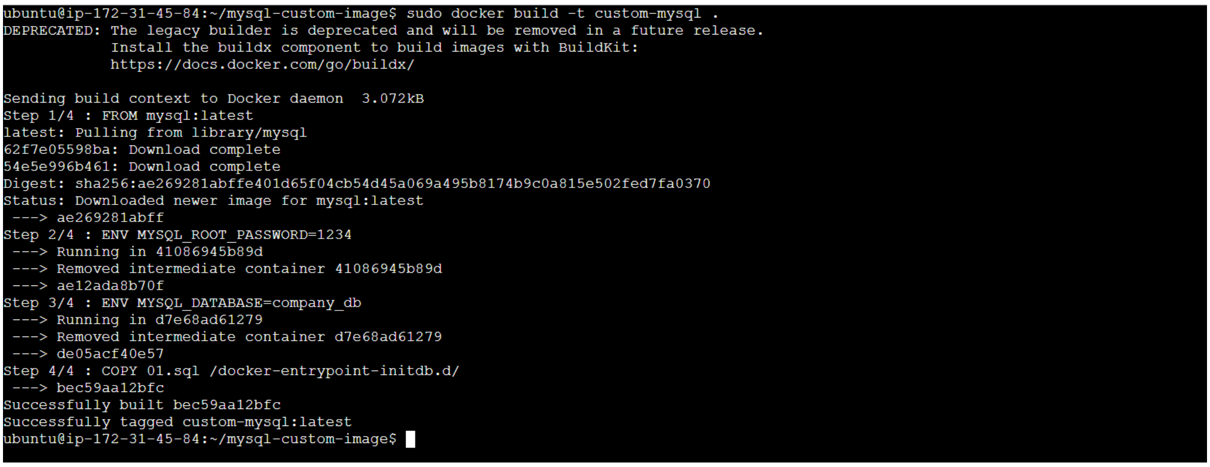
```
FLUSH PRIVILEGES;
```

- Save the file.
- Create a Dockerfile.
- nano Dockerfile
- Add:

```
FROM mysql:latest

ENV MYSQL_ROOT_PASSWORD=1234

ENV MYSQL_DATABASE=company_db

COPY 01.sql /docker-entrypoint-initdb.d/
```

- Save the file.
- **Build the Custom Image**

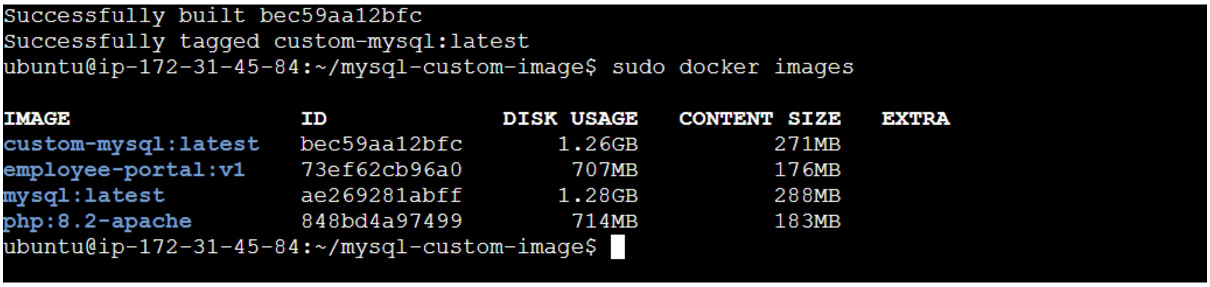
```
sudo docker build -t custom-mysql .
```



```
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker build -t custom-mysql .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.072kB
Step 1/4 : FROM mysql:latest
latest: Pulling from library/mysql
62f7e05598ba: Download complete
54e5e996b461: Download complete
Digest: sha256:ae269281abffe401d65f04cb54d45a069a495b8174b9c0a815e502fed7fa0370
Status: Downloaded newer image for mysql:latest
 ---> ae269281abff
Step 2/4 : ENV MYSQL_ROOT_PASSWORD=1234
 ---> Running in 41086945b89d
 ---> Removed intermediate container 41086945b89d
 ---> ae12ada8b70f
Step 3/4 : ENV MYSQL_DATABASE=company_db
 ---> Running in d7e68ad61279
 ---> Removed intermediate container d7e68ad61279
 ---> de05acf40e57
Step 4/4 : COPY 01.sql /docker-entrypoint-initdb.d/
 ---> bec59aa12bfc
Successfully built bec59aa12bfc
Successfully tagged custom-mysql:latest
ubuntu@ip-172-31-45-84:~/mysql-custom-image$
```

- **Verify the Image**

```
sudo docker images
```



```
Successfully built bec59aa12bfc
Successfully tagged custom-mysql:latest
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker images

IMAGE                 ID             DISK USAGE   CONTENT SIZE   EXTRA
custom-mysql:latest   bec59aa12bfc       1.26GB          271MB
employee-portal:v1    73ef62cb96a0        707MB          176MB
mysql:latest          ae269281abff       1.28GB          288MB
php:8.2-apache        848bd4a97499        714MB          183MB
ubuntu@ip-172-31-45-84:~/mysql-custom-image$
```

- Stop and Remove Any Existing MySQL Container
- Run a Container from the Custom Image

```
sudo docker run -d --name mysql-instance -p 3306:3306 custom-mysql
```

```
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker run -d --name mysql-instance -p 3306:3306 custom-mysql
5165feb5a40592db6f4ce570e8483d4cfdc0798b9b6454c743d993f67cf1ed2b
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                                                    NAMES
5165feb5a405   custom-mysql   "docker-entrypoint.s…"   22 seconds ago   Up 21 seconds   0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp   mysql-instance
21465fd39faf   30b895db7822   "docker-entrypoint.s…"   8 minutes ago    Created                                                                  serene_lovelace
ubuntu@ip-172-31-45-84:~/mysql-custom-image$
```

- **Connect to MySQL**

```
sudo docker exec -it mysql-instance mysql -u root -p
```

- Password
- **Select the Database**

```
USE company_db;
```

- **Verify the Data**

```
SELECT * FROM employees;
```

```
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker exec -it mysql-instance mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 9.7.1 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE company_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM employees;
+-------------+---------------+------------+----------+
| employee_id | employee_name | department | salary   |
+-------------+---------------+------------+----------+
|           1 | Aman          | Cloud      | 65000.00 |
|           2 | Rahul         | DevOps     | 72000.00 |
|           3 | Priya         | AI/ML      | 85000.00 |
+-------------+---------------+------------+----------+
3 rows in set (0.003 sec)

mysql>
```

Persist MySQL Data Using Docker Volumes

- List all Docker volumes.

```
sudo docker volume ls
```

```
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker volume ls
DRIVER    VOLUME NAME
local     1c28e9a2819f187dbadacfd25e89c4a9a2f7245787a26ae7e9cc773e4ff06bca
local     100cadc03c80c3130338b48c8ceb0905071f82ed34849626af1fa679b1717e83
local     c988424d7f110d313b5e0dc54da2733264b463e6ebb61ff91defd55c1ad5b3dd
local     dbf2f23142bd26cd37fcbd8caf4208e9841da92f889e3e53b30fa78aaa63acb3
ubuntu@ip-172-31-45-84:~/mysql-custom-image$
```

- Create a volume to store MySQL data.

```
sudo docker volume create mysql-data
```

- Verify the volume.

```
sudo docker volume ls
```

```
ubuntu@ip-172-31-45-84:~/mysql-custom-image$
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker volume create mysql-data
mysql-data
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker volume ls
DRIVER    VOLUME NAME
local     1c28e9a2819f187dbadacfd25e89c4a9a2f7245787a26ae7e9cc773e4ff06bca
local     100cadc03c80c3130338b48c8ceb0905071f82ed34849626af1fa679b1717e83
local     c988424d7f110d313b5e0dc54da2733264b463e6ebb61ff91defd55c1ad5b3dd
local     dbf2f23142bd26cd37fcbd8caf4208e9841da92f889e3e53b30fa78aaa63acb3
local     mysql-data
ubuntu@ip-172-31-45-84:~/mysql-custom-image$
```

- Stop and Remove the Existing MySQL Container
- Create a new container and attach the volume.

```
sudo docker run -d --name mysql-instance -p 3306:3306 -v mysql-data:/var/lib/mysql
custom-mysql
```

Log in as the MySQL root user inside the container

```
sudo docker exec -it mysql-instance mysql -u root -p
```

- **Verify Existing Data**

```
USE company_db;

SELECT * FROM employees;
```

- Insert a New Record

```
INSERT INTO employees

(employee_name, department, salary)

VALUES

('Neha','Cyber Security',78000);
```

```
mysql> INSERT INTO employees
    -> (employee_name, department, salary)
    -> VALUES
    -> ('Neha','Cyber Security',78000);
Query OK, 1 row affected (0.008 sec)

mysql> SELECT * FROM employees;
+-------------+---------------+----------------+----------+
| employee_id | employee_name | department     | salary   |
+-------------+---------------+----------------+----------+
|           1 | Aman          | Cloud          | 65000.00 |
|           2 | Rahul         | DevOps         | 72000.00 |
|           3 | Priya         | AI/ML          | 85000.00 |
|           4 | Neha          | Cyber Security | 78000.00 |
+-------------+---------------+----------------+----------+
4 rows in set (0.000 sec)

mysql>
```

- **Stop and Remove the Container**

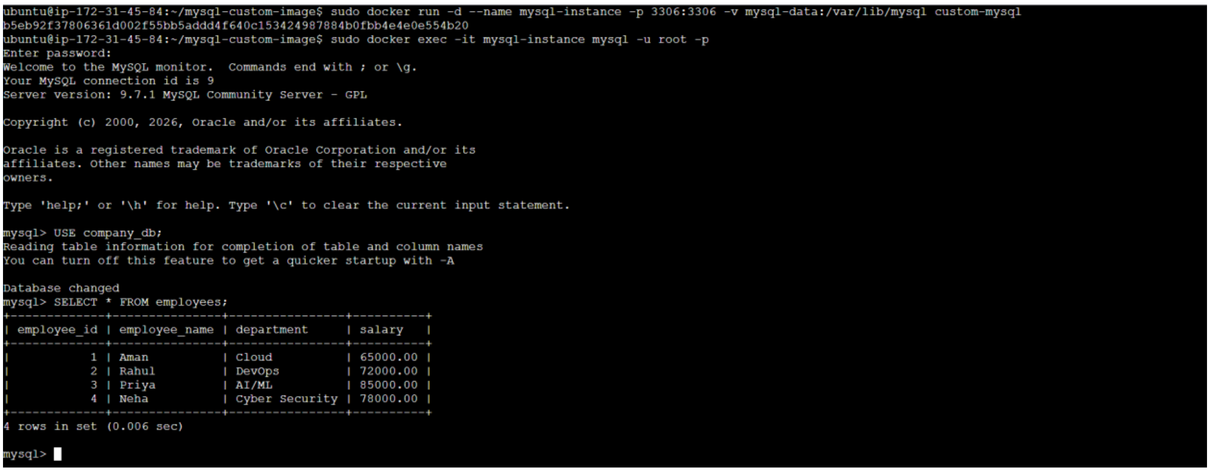
```
sudo docker stop mysql-instance

sudo docker rm mysql-instance
```

- **Recreate the Container Using the Same Volume**

```
sudo docker run -d --name mysql-instance -p 3306:3306 -v mysql-data:/var/lib/mysql
custom-mysql
```

Reconnect to MySQL.

```
sudo docker exec -it mysql-instance mysql -u root -p
```



```
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker run -d --name mysql-instance -p 3306:3306 -v mysql-data:/var/lib/mysql custom-mysql
b5eb92f37806361d002f55bb5addd4f640c153424987884b0fbb4e4e0e554b20
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker exec -it mysql-instance mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 9.7.1 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE company_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM employees;
+-------------+---------------+----------------+----------+
| employee_id | employee_name | department     | salary   |
+-------------+---------------+----------------+----------+
|           1 | Aman          | Cloud          | 65000.00 |
|           2 | Rahul         | DevOps         | 72000.00 |
|           3 | Priya         | AI/ML          | 85000.00 |
|           4 | Neha          | Cyber Security | 78000.00 |
+-------------+---------------+----------------+----------+
4 rows in set (0.006 sec)

mysql>
```
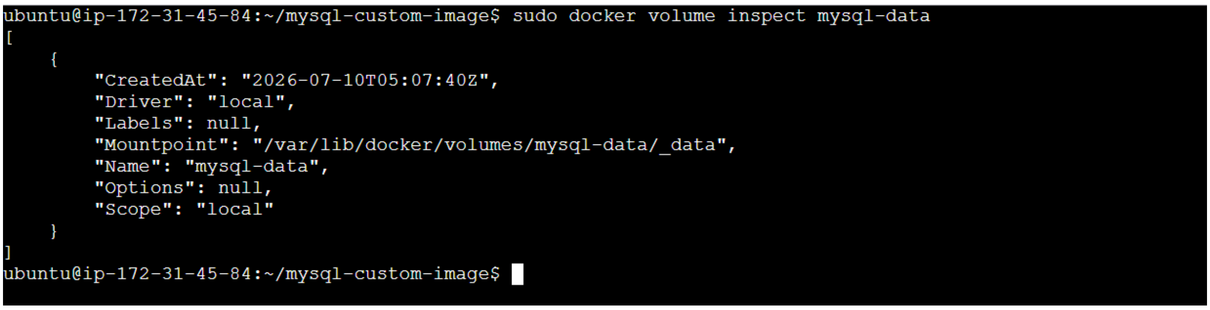

- The previously inserted record (Neha) should still exist, confirming that the data was preserved by the Docker volume.
- Exit MySQL.

```
EXIT;
```

- Inspect the volume.

```
sudo docker volume inspect mysql-data
```



```
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker volume inspect mysql-data
[
    {
        "CreatedAt": "2026-07-10T05:07:40Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/mysql-data/_data",
        "Name": "mysql-data",
        "Options": null,
        "Scope": "local"
    }
]
ubuntu@ip-172-31-45-84:~/mysql-custom-image$
```

- Prepare a Custom MySQL Image for a New Application
- Edit a SQL file 01.sql

```
CREATE DATABASE IF NOT EXISTS company_db;

USE company_db;

-- Create employees table

CREATE TABLE employees (
```

```
    employee_id INT AUTO_INCREMENT PRIMARY KEY,
    employee_name VARCHAR(100),
    department VARCHAR(50),
    salary DECIMAL(10,2),
    email VARCHAR(100)
);
-- Insert sample data
INSERT INTO employees (employee_name, department, salary, email)
VALUES
('Aman', 'Cloud', 65000, 'aman@example.com'),
('Rahul', 'DevOps', 72000, 'rahul@example.com'),
('Priya', 'AI/ML', 85000, 'priya@example.com');
-- Create a new MySQL user
CREATE USER 'appuser'@'%' IDENTIFIED BY 'AppUser@123';
-- Grant privileges on the database
GRANT ALL PRIVILEGES ON company_db.* TO 'appuser'@'%';
-- Apply the privilege changes
FLUSH PRIVILEGES;
-- Verify the data
SELECT * FROM employees;
```

- **Check Running Containers**

```
sudo docker ps
```

- Stop the running container.

```
sudo docker stop mysql-instance
```

- Remove it.

```
sudo docker rm mysql-instance
```

- Build a New Custom MySQL Image

```
sudo docker build -t mysql-exam-image .
```

- **Verify the Image**

```
sudo docker images
```

```
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker rm mysql-instance
mysql-instance
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker build -t mysql-exam-image .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.584kB
Step 1/4 : FROM mysql:latest
 ---> ae269281abff
Step 2/4 : ENV MYSQL_ROOT_PASSWORD=1234
 ---> Using cache
 ---> ae12ada8b70f
Step 3/4 : ENV MYSQL_DATABASE=company_db
 ---> Using cache
 ---> de05acf40e57
Step 4/4 : COPY 01.sql /docker-entrypoint-initdb.d/
 ---> 2605530a8799
Successfully built 2605530a8799
Successfully tagged mysql-exam-image:latest
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker images

IMAGE                     ID             DISK USAGE   CONTENT SIZE   EXTRA
custom-mysql:latest       bec59aa12bfc       1.26GB          271MB
employee-portal:v1        73ef62cb96a0        707MB          176MB
mysql-exam-image:latest   2605530a8799       1.26GB          271MB
mysql:latest              ae269281abff       1.28GB          288MB
php:8.2-apache            848bd4a97499        714MB          183MB
ubuntu@ip-172-31-45-84:~/mysql-custom-image$
```

- **List Existing Docker Volumes**

```
sudo docker volume ls
```

- **Remove the Existing Docker Volume**

```
sudo docker volume rm mysql-data
```

- **Create a New Docker Volume**

```
sudo docker volume create mysql-data
```

- Verify it.

```
sudo docker volume ls
```

```
ubuntu@ip-172-31-45-84:~/mysql-custom-image$
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker volume ls
DRIVER    VOLUME NAME
local     1c28e9a2819f187dbadacfd25e89c4a9a2f7245787a26ae7e9cc773e4ff06bca
local     100cadc03c80c3130338b48c8ceb0905071f82ed34849626af1fa679b1717e83
local     c988424d7f110d313b5e0dc54da2733264b463e6ebb61ff91defd55c1ad5b3dd
local     dbf2f23142bd26cd37fcbd8caf4208e9841da92f889e3e53b30fa78aaa63acb3
local     mysql-data
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker volume rm mysql-data
mysql-data
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker volume create mysql-data
mysql-data
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker volume create mysql-data
mysql-data
ubuntu@ip-172-31-45-84:~/mysql-custom-image$
```

Run the New MySQL Container

- Create a container using the new image and attach the new volume.

```
sudo docker run -d --name mysql-instance -p 3306:3306 -v mysql-data:/var/lib/mysql mysql-exam-image
```

- **Verify the Running Container**

```
sudo docker ps
```

```
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker run -d --name mysql-instance -p 3306:3306 -v mysql-data:/var/lib/mysql mysql-exam-image
65b755187301c50ff11fee6bccc7e2944c244ed2975c24bc1109b1de8c7ec858
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED         STATUS         PORTS                                                    NAMES
65b755187301   mysql-exam-image  "docker-entrypoint.s…"   8 seconds ago   Up 7 seconds   0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp   mysql-instance
ubuntu@ip-172-31-45-84:~/mysql-custom-image$
```

- Connect to MySQL

```
sudo docker exec -it mysql-instance mysql -u root -p
```

- Select the database.

```
USE company_db;
```

- View the data.

```
SELECT * FROM employees;
```

- Exit MySQL.

```
EXIT;
```

```
ubuntu@ip-172-31-45-84:~/mysql-custom-image$ sudo docker exec -it mysql-instance mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 9.7.1 MySQL Community Server - GPL

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mysql> USE company_db;
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Database changed
mysql> SELECT * FROM employees;
+-------------+---------------+------------+----------+-------------------+
| employee_id | employee_name | department | salary   | email             |
+-------------+---------------+------------+----------+-------------------+
|           1 | Aman          | Cloud      | 65000.00 | aman@example.com  |
|           2 | Rahul         | DevOps     | 72000.00 | rahul@example.com |
|           3 | Priya         | AI/ML      | 85000.00 | priya@example.com |
+-------------+---------------+------------+----------+-------------------+
3 rows in set (0.000 sec)

mysql> exit
Bye
ubuntu@ip-172-31-45-84:~/mysql-custom-image$
```